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Stability of multi-vendor power electronicsdominated grids requires proper specifications for converters interoperability, e.g. tuning boundaries for the converters' control loops to be respected by all vendors. The confidentiality and variety of implementation of the converters' control system by different vendors and the nonlinear dynamics of power converters make this task arduous. This paper formulates this research question as a robust stability problem, embedding frequency-dependent uncertainties in the converter control systems and analytically quantifying the amount of uncertainty able to warranty the power grid stability. Grid following converters are considered and modeled with a nonlinear grey-box approach, in contrast with conventional small-signal black-box models which suffer from inaccuracy when their operating point deviates from the nominal one. This paper also demonstrates both analytically and with Hardware-In-the-Loop (HIL) experiments that different converters operating points (e.g. active power injections, ac bus voltages) radically affect their stability, empathizing the necessity of nonlinear converters models to realize accurate power grid stability analyses.

Keywords

converter-based renewable, differential-algebraic equations, grey-box modeling, interoperability, microgrids, nonlinear state-space model, power electronics, power systems, power, energy and industry applications, robust control, stability analysis, structured singular values

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Interoperability Specifications for Multi-Vendor Converter-Dominated Grid: a Robust Stability Perspective

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***Abstract*—Stability of multi-vendor power electronics-dominated grids requires proper specifications for converters interoperability, e.g. tuning boundaries for the converters' control loops to be respected by all vendors. The confidentiality and variety of implementation of the converters' control system by different vendors and the nonlinear dynamics of power converters make this task arduous. This paper formulates this research question as a robust stability problem, embedding frequency-dependent uncertainties in the converter control systems and analytically quantifying the amount of uncertainty able to warranty the power grid stability. Grid following converters are considered and modeled with a nonlinear grey-box approach, in contrast with conventional small-signal black-box models which suffer from inaccuracy when their operating point deviates from the nominal one. This paper also demonstrates both analytically and with Hardware-In-the-Loop (HIL) experiments that different converters operating points (e.g. active power injections, ac bus voltages) radically affect their stability, empathizing the necessity of nonlinear converters models to realize accurate power grid stability analyses.**

***Index Terms*—Interoperability, Grey-box modeling, Nonlinear state-space model, Differential-Algebraic Equations, Stability analysis, Robust control, Grid Following Converters, Structured Singular Values, H-infinity**

I. Introduction

Stability analyses of power electronics-dominated power systems require accurate converters models, whose derivation has been object of intensive research in the last decades [1]–[3].

Linear models of Voltage Source Converters (VSCs) are nowadays widely used for power systems small-signal stability analyses, both defined in the frequency domain as impedance-based models [4], [5] and in the time-domain as state-space models [3], [6], [7]. State-space grid models are favourable for system-level oscillations investigation and participation factor analyses, yet they are conventionally constructed by assuming the detailed control structure and parameters of power converters fully known [6]. In many cases this assumption is not realistic, since the detailed implementation of the control system is kept confidential by the industries, to protect the intellectual property (IP) [8].

Due to the IP-protection, impedance-based black-box converter models estimated from measurements increased in popularity, and a great effort has been made toward more sophisticated dq and $\alpha\beta$ impedance estimation techniques [9]–[11] and frequency-domain power system stability analysis methodologies [4], [12]–[14]. The paper [15] analysed the problem of converter interoperability in multi-terminal HVDC grids through impedance-based models and sensitivity analysis. While the analyses in [15] are useful to identify which converter is causing the instability, they are not aimed to provide precise control guidelines (e.g. tuning boundaries for the control loops) to avoid instability. The use of impedance-based converter models for stability and interoperability analysis in modern power system analysis, despite their black-box modeling capability, have also another remarkable limitation.

Conventional impedance models identified through measurement are *small-signal*, thus accurate only around the nominal operating point considered in the measurement process [16]. Due to the high variability of power injection by renewables and load demand profiles, modern grids operate in a wide range of steady-state conditions, where converter small-signal models may be inaccurate [3], [8]. Recent studies confirmed indeed that the stability margin of PLL-synchronized converters operating with low ac bus voltage levels can be significantly lower than in nominal conditions [17]. The injection of reactive current, instead, has been found to be beneficial for the stability in PLL-synchronized converters [18], yet detrimental for the stability in power-synchronized converters [19].

The aforementioned limitations sharpened the inadequacies of small-signal converter impedance models in power grid stability analyses and motivated the investigation of new methodologies based on *nonlinear* converter models, that allow analyses under all grid operating conditions. To this end, our work [3] developed nonlinear models of power converters and used them to build nonlinear Differential-Algebraic Equations (DAE) models of converter-dominated grids. Nonlinear DAE models are ubiquitous in the traditional machine-dominated grids literature and have been employed for power flow analyses [20], transient stability analyses [21], time-domain simulations, continuation power flow, optimal power flow [22] and voltage stability analyses [23].

The derivation and use of nonlinear DAE models for converter-dominated grids, although promising, must necessarily take into account the different characteristics of power electronics-based grids compared to traditional power systems. Firstly, the reduced inertia and the faster timescales of Inverter-Based Resources (IBR) with respect to Synchronous Generators (SG) have several implications on the mathematical

modeling of the grid, including the inadequacy of the quasi-sinusoidal state assumption and the phasor approximation [1], and the need for accurately modeling the IBR control system in a sufficiently fast timescale, coupled with the electromagnetic transient dynamics of the power network [4]. Secondly, the IP-protection of the converter control system is a major obstacle for nonlinear DAE and/or state-space modeling of power converters. While the converter-dominated grid DAE model developed in [3] correctly models the fast-acting control system of the IBR, it assumes full knowledge of the converter control systems and is therefore not suitable for interoperability and stability analyses in the presence of IP-protected converter models.

This paper aims to overcome these challenges by adopting a robust stability approach, i.e. embedding structured perturbations in the converters models to represent the uncertainty of the control system. The main novelties and key findings of the proposed approach are here summarized:

- The converter modeling strategy follows a nonlinear grey-box paradigm, in contrast with conventional linear black-box impedance models, assuming only a rough and non-detailed knowledge of the control system (e.g. if grid forming or grid following).
- The robust control theory (i.e. μ-analysis) is applied to analytically derive the *interoperability specifications* for the converters, in the form of tuning boundaries for the control loops that, if respected by all vendors, can warranty the grid stability.
- Advanced analyses such as destabilizing perturbation analysis and sensitivity analysis are applied to detect hazardous control tuning and consequently find the interoperability specifications for the converters.
- Differently from conventional impedance models, the proposed converter model is nonlinear and accurately predict the stability margin in all grid operating conditions, detecting potentially hazardous grid operating points. This paper demonstrates that, depending on the converters and power system equilibrium point, the stability margin can significantly vary, highlighting the potential inaccuracy of small-signal converter models in power system analyses.

The paper is structured as follows: section II gives a theoretical background, section III models the control loops, section IV introduces the converter and grid model, section V performs robust stability analyses, section VI finds the interoperability specifications, section VII shows the Hardware In the Loop (HIL) results, and section VIII the conclusions.

II. Modeling of Grid-Connected Converters with Uncertain Control System

Traditional impedance-based models estimated from measurements consider the converter dynamics as a *linear black-box*, making little to no assumption on the converter hardware and control structure [8], [9]. However, due to the nonlinearity of the converter dynamics, small-signal converter models may introduce inaccuracy in power system analyses, motivating the investigation of nonlinear models [3], [8], [18], [30], [31]. For the nonlinear converter modeling, despite some specific details of the control implementation remain unknown, the knowledge of power converter physical laws and basic operation in power grids can give enough information to assume it a *nonlinear grey-box*, rather than a black-box.

Table I provides an overview on the dynamics of grid-connected converters, separating the ones which can be realistically assumed to be known, and the ones that, due to IP-protections by converter manufacturers, are obscure to system operators. Robust control theory provides a mathematical framework to model unknown dynamics (as the ones described in Table I) by means of structured uncertainties, and to analyse the system stability margin with respect to the magnitude of the uncertainty. Its application to the determination of the *interoperability specifications* of grid-connected converters represents, in the best of the authors' knowledge, a unique novelty of this paper.

The grey-box models presented in this section are applicable only to represent *grid following* converters which adopt the generic control scheme depicted in Fig. 1(a). Grid forming converters need a different definition of the grey-box model structure, which is outside the scope of this work. The mathematical methodology for interoperability analysis discussed in section V and VI is, however, easily extendable also power grids with the presence of grid forming converters.

A. Grey-box nonlinear modeling of grid-following converter

A multi-vendor power grid with three grid following (GFL) VSCs with unknown control system, as in Fig. 1(b), is considered. A general representation of the GFL control adopted in three-phase converters is illustrated in Fig. 1(a). The generic control structure composed of an inner current control $G_{ci}(s)$, a dc voltage control $G_{dvc}(s)$, a voltage-based synchronization $G_{syn}(s)$ (e.g. PLL) and and ac voltage control loop $G_{avc}(s)$. Due to the IP-protection, the transfer functions $G_{ci}(s)$, $G_{dvc}(s)$, $G_{syn}(s)$, $G_{avc}(s)$ are unknown for system operators, both in their parameters and in their order, and thus depicted in grey colour in Fig. 1(a). In case of grid following converters with slightly different control structure, e.g. reactive power control instead of ac voltage control, the grey-box structure in Fig. 1(a) has to be adjusted accordingly, without loss of generality of the following proposed methodology. The converter nonlinearity must be modeled to properly represent the operating point-dependent behaviour of the converter for system-level stability analyses.

The proposed nonlinear modeling strategy for the converter in Fig. 1(a) is graphically explained in Fig. 2. The model is composed of both nonlinear blocks (in red) containing nonlinear input-output differential-algebraic operators, and linear blocks (in green) containing transfer functions. The dynamics of the dc side of the converter is governed by the power conservation law nonlinear DAE:

$$\begin{cases} \dot{v}_{dc} &= -\dfrac{3}{2}\dfrac{1}{C_{dc}}\dfrac{v_{g_{dq}}^T i_{g_{dq}}}{v_{dc}} + \dfrac{1}{C_{dc}} i_{dc} \\ y &= v_{dc} \end{cases} \tag{1}$$

TABLE I: Overview of the known and unknown dynamics and information in grid-connected power converters, which focus on grid following converters.

Known Dynamics	**Comment**
Converter operation mode, e.g. grid following or grid forming	The operation mode of power converters is not confidential and generally known.
Grid following general control structure	A solid consensus in the industrial and scientific community exists regarding grid following converters implementation, using a PLL to track the grid voltage angle and regulating active and reactive power by means of direct and quadrature component of the current [24], [25]. The dc-link voltage is regulated by means of the direct current.
Control tuning for grid supporting functionalities	Grid codes impose specific requirements for the injection of reactive current/power depending on the grid voltage deviation [26]. The slope of the curve is generally declared in the grid code, giving relevant information for the converter modeling.
Known equations in the power converters model related to fundamental laws of electromagnetism and well-known mathematical relationships	- Energy conservation law between ac and dc side, dc-link capacitor equation (1) - Rotation matrices T_δ in (3) for transforming between local and global reference frame - Integral relationship between frequency and angle, i.e. $\dot{\delta} = \omega$
IP-Protected and Unknown Dynamics	**Comment**
PLL implementation, type and tuning	There is no consensus in the industrial and scientific community about how to tune the PLL. Moreover, there exist many PLL implementations involving filters and/or sequence decoupling networks to increase the phase tracking performances, including Moving Average Filters, Notch Filters, Decoupled Double Synchronous Reference Frame PLL [25], [27], [28].
DC-link voltage control tuning	There is no full consensus in the industrial and scientific community about the tuning of the dc voltage control. Also, different implementation such as linear and quadratic voltage control exist [29].
Current controller	Techniques for tuning the current loop are well-established, however different implementation exist (e.g. PI, PR). Furthermore, negative sequence and harmonic compensation schemes can be implemented in the current loop [25], [30].

assuming no conversion losses. The currents and voltages are expressed as dq vectors, e.g. $v_{g_{dq}} = \begin{pmatrix} v_{g_d} \\ v_{g_q} \end{pmatrix}$. The subscript dq is referred to the variable expressed in the local dq reference frame of the converter, and the subscript DQ for the global grid frame, as explained in [3]. For the rest of the paper, the notation $v_g := v_{g_{dq}}$ is adopted to increase the readability. Equation (1) is expressed in compact form in Fig. 2 as:

$$\begin{cases} \dot{x} = f_{dc}(x, u) \\ y = h_{dc}(x, u) \end{cases} \tag{2}$$

Under the assumption of known size of the dc capacitor C_{dc}, (1) can be assumed know without uncertainty. In the case where C_{dc} is unknown, it can be easily estimated by exciting the dc converter side with small voltages while the converter is not switching. The converter model in Fig. 2 presents both input/output for the the ac grid interconnection (v_{gDQ}, i_{gDQ}) and dc interconnection (v_{dc}, i_{dc}), according to [3].

The other nonlinear blocks in Fig. 2 are the direct and inverse rotation matrices $T(\delta)$ and $T^{-1}(\delta)$, used to transform the three-phase variables from local dq frame into global DQ frame [3]. The matrix $T(\delta)$ is defined as:

$$T(\delta) = \begin{pmatrix} \cos\delta & \sin\delta \\ -\sin\delta & \cos\delta \end{pmatrix} \tag{3}$$

and used in the algebraic relations $i_{g_{dq}} = T(\delta) i_{g_{DQ}}$ and $v_{g_{DQ}} = T^{-1}(\delta) v_{g_{dq}}$. Clearly, there is no uncertainty on the matrices $T(\delta)$ and $T^{-1}(\delta)$, which can thus be considered as known blocks in the converter model in Fig. 2.

The remaining blocks in Fig. 2 are all linear: the integral relationship $\dot{\delta} = \omega$ is linear and certain, while the transfer functions $G_{ci}(s)$, $G_{dvc}(s)$, $G_{syn}(s)$ and $G_{avc}(s)$ are linear but unknown. In the light of the previous discussion and the representation in Fig. 2, an assumption of extreme importance to justify the proposed methodology is drawn.

Assumption:. *Assuming that the converter control loops can be modeled as linear transfer functions (as done in most of the actual literature), the nonlinear dynamics* (1) *and* (3) *of the converter model can be separated from the linear control system model. The unknown dynamics of the converter control system are confined inside the dashed green frame in Fig. 2, and are linear. The nonlinear dynamics* (1) *and* (3) *represented in red rectangles in Fig. 2 lay outside the dashed green frame, and are not affected by uncertainty.*

The uncertain converter control system (green dashed frame in Fig. 2) can be thus modeled by means of linear multiplicative perturbations defined in the frequency-domain on the single control loops [32]. In other words, each of the control loops in Fig. 2 is defined as unknown transfer functions $G_{ci}(s)$, $G_{dvc}(s)$, $G_{syn}(s)$, $G_{avc}(s)$, whose frequency response lays in a specific region of the Bode plot (see Fig. 3).

B. Application of robust control theory to interoperability analysis

Robust control theory is commonly employed to quantify how much uncertainty can a dynamical system tolerate before it becomes unstable. In the specific context of converter-dominated grids, where the uncertainty encompasses all different and confidential approaches of design and tuning of the converters' control loops by diverse vendors, robust stability analysis can actually determine which are the boundaries for the control system tuning which ensure the power grid stability. Such tuning boundaries have the form of a region in the Bode plots where the transfer functions $G_{ci}(s)$, $G_{dvc}(s)$, $G_{syn}(s)$, $G_{avc}(s)$ must lay (see white regions in Fig. 3). If these tuning boundaries can ensure grid robust stability, then they constitute the *interoperability specifications* for power converters, i.e. a set of tuning recommendations and guidelines for converter manufacturers defined by the system operator for

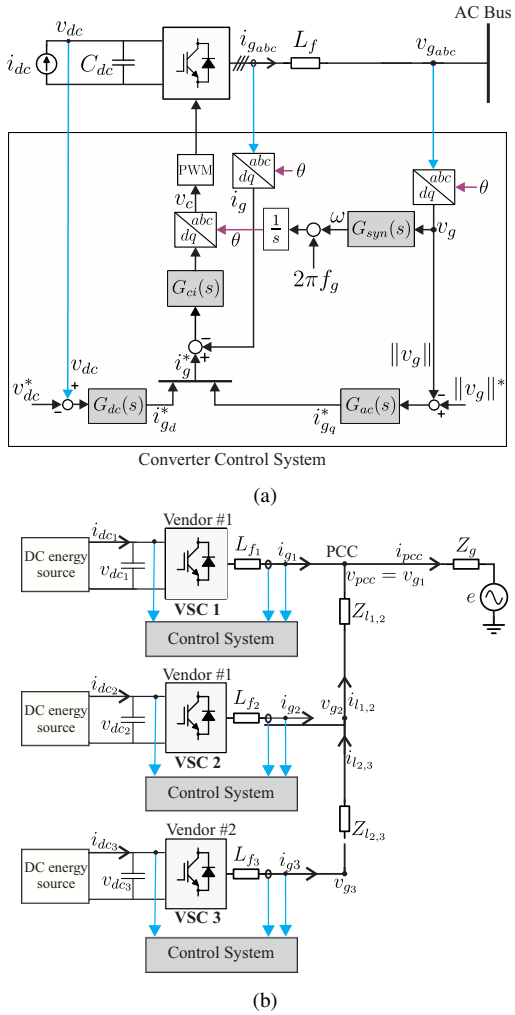


Fig. 1: (a) The block scheme of the considered grid following converter with uncertain control system represented by grey blocks. (b) The considered power grid with three grid following converters from different vendors with unknown control system.

ensuring safe and stable operation of the grid. The smaller the tuning boundaries (i.e. lower uncertainty), the stricter the guidelines imposed to the converter vendors. The larger the boundaries (i.e. higher uncertainty), the wider the freedom of control tuning and implementation permitted to the vendors, which can still maintain the IP-protection of their control system details.

The effective application of this concept necessitate a proper and targeted definition of uncertainties, i.e. a proper modeling of the unknown grey transfer functions in Fig. 2. For that, a frequency-domain multiplicative uncertainty is employed for each of the control loops $G_{ci}(s)$, $G_{dvc}(s)$, $G_{syn}(s)$, $G_{avc}(s)$, formulated in a generic form as [32]:

$$G(s) = \bar{G}(s)(1 + w\Delta(s)) \tag{4}$$

The transfer function $G(s)$ represents the *real plant* of a linear dynamic system (i.e. the control loop). The uncertain formulation in (4) includes:

- The transfer function $\bar{G}(s)$ of the *nominal* control loop implementation, generally chosen according to an *initial guess*. For that, a simple and conventional implementation (e.g. proportional or PI controller) is generally assumed, with the control gains tuned according to well-consolidated techniques in literature [25].
- The system uncertainty $\Delta(s)$, which describes the *frequency-dependent* deviation of the real plant $G(s)$ from its initial guess $\bar{G}(s)$. The initial design of $\Delta(s)$ is based on prior intuitions regarding the control loop. Note that no specifications are imposed regarding the order of $\Delta(s)$, which can be also a high-order transfer function modeling a complex implementation of the control loop. If a higher uncertainty is expected in the high frequency range, for example, $\Delta(s)$ is chosen as a high-pass filter. The Matlab® commands used for defining $\Delta(s)$ are `ultidyn()` to create an unknown transfer function with unitary $\mathcal{H}_\infty$ norm, and `makeweight()` to shape its frequency response.
- The scalar constant w, which quantifies the amount of uncertainty. Considering the formulation in (4) and that $\|\Delta(s)\|_\infty = 1$, w represents the maximum deviation of the transfer function $G(s)$ with respect to its nominal value $\bar{G}(s)$, in a $\mathcal{H}_\infty$ norm sense.

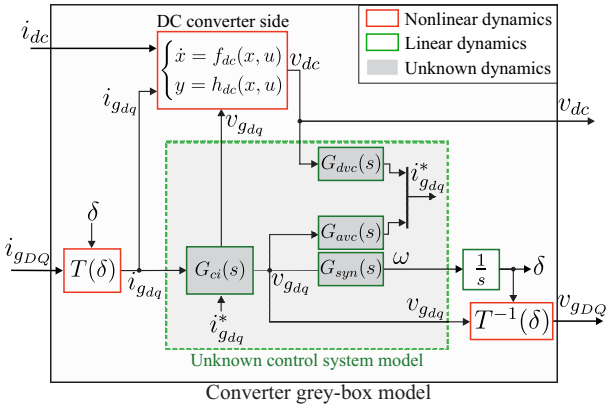


Fig. 2: The proposed modeling scheme for the considered converter including linear, nonlinear and uncertain blocks.

The determination of the the correct formulation (4) for each control loop is conducted as follows. An initial definition of the uncertainties for each control loop is given based on prior intuitions on the the control system (see section III). Afterward, the power grid model with the proposed uncertainty is build (see section IV) and studied by means of robust stability analysis (see section V). In the case where the power grid is *not* robustly stable, the control loops uncertainties are reformulated by adjusting and properly restricting w and/or $\Delta(s)$ until they can warranty the grid stability, according to the procedure in section VI. At the end of this procedure, the *interoperability specifications* for the power converters in the considered grid are obtained.

III. Quantitative Initial Design of the Uncertainty in the Converter Control System

The formulation for the uncertainty in (4) is applied in this section to the four uncertain control loops $G_{ci}(s)$, $G_{dvc}(s)$, $G_{syn}(s)$ and $G_{avc}(s)$, shown in Fig. 2. The initial definition

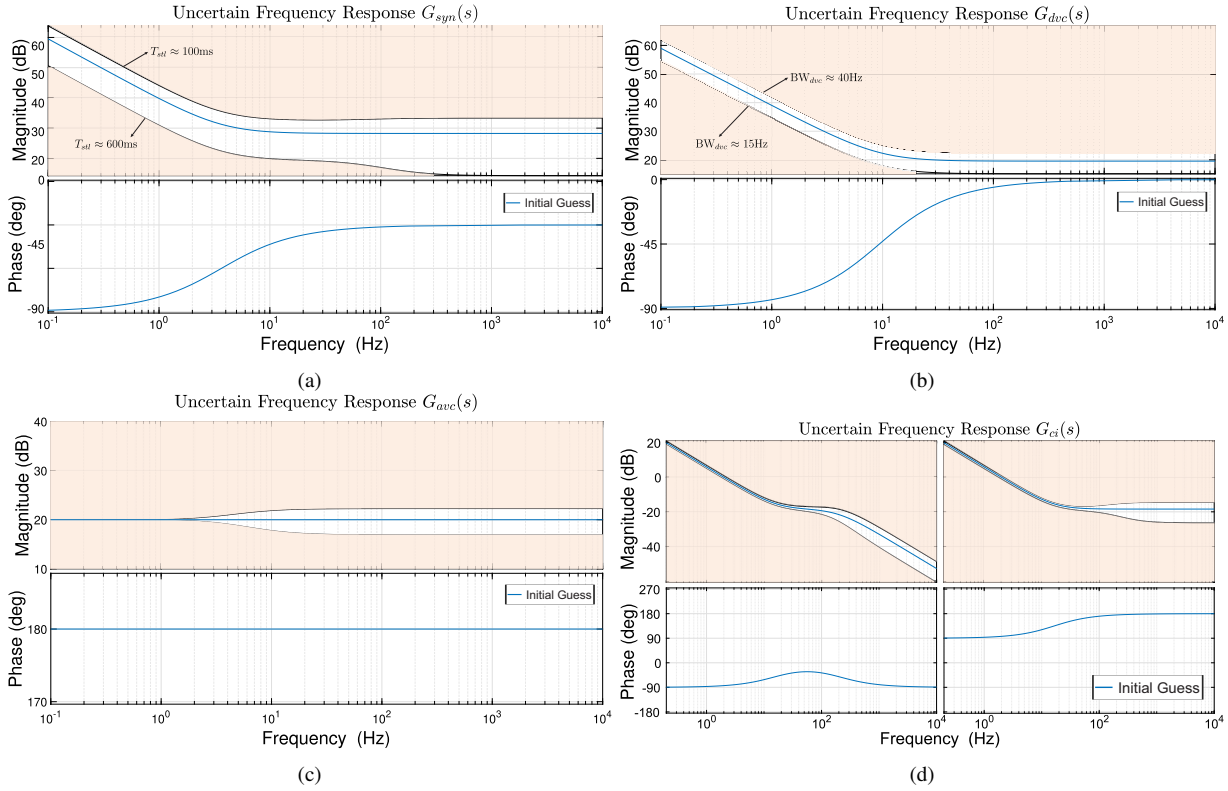


Fig. 3: The uncertain frequency response of the individual control loops, laying inside the white region of the Bode plot. (a) The synchronization loop. (b) The dc voltage loop. (c) The ac voltage loop. (d) The current loop.

of the nominal transfer function $\bar{G}(s)$ and the uncertainty $\Delta(s)$ is done loop-by-loop, tailored to the individual characteristics and possible implementations of each specific loop.

A. Synchronization loop $G_{syn}(s)$

The synchronization in three-phase grid following converters is realized, in most of the cases, by means of a Phase-Locked Loop (PLL) [25]. The Synchronous Reference Frame PLL (SRF-PLL) is a widely PLL implementation, nevertheless it is not the only one; several other implementations involving sequence detectors or harmonic rejection algorithms have been proposed in literature [25], [27]. Fig. 4 shows, for example, a particular type of PLL with sequence decoupling network, denominated Decoupled Double Synchronous Reference Frame PLL (DDSRF-PLL), which will be implemented in the HIL real-time simulations in section VII for proving the generality of the proposed method [28]. The PLL specific implementation, as well as its tuning (e.g. settling time), are generally kept confidential by manufacturers [25]. In such cases, a nominal plant (i.e. initial guess) $\bar{G}_{syn}(s)$ for the synchronization loop represented in grey colour in Figs. 1(a) and 2 is firstly defined, and the unknown dynamics are then included through a proper frequency-dependent uncertainties $w_{syn}\Delta_{syn}(s)$.

For a high generalizability of the proposed procedure, the nominal plant $\bar{G}_{syn}(s)$ has to be chosen as simple and possible, such that more complex plants can be properly encompassed by the defined uncertainty. An SRF-PLL with a PI controller is thus assumed, acting on the q component of v_g and producing the frequency ω at the output:

$$\begin{cases} \omega & = \bar{G}_{syn}(s) v_{g_q} \\ \bar{G}_{syn}(s) & = \dfrac{K_{p,PLL}\, s + K_{i,PLL}}{s} \end{cases} \tag{5}$$

The tuning of the PLL gains $K_{p,PLL}$,$K_{i,PLL}$ is usually realized to obtain optimal dynamic damping with a desired settling time T_{stl}, e.g. $K_{p,PLL} = \frac{9.2}{T_{stl}}$ and $K_{i,PLL} = 0.21 K_{p,PLL} T_{stl}$. Being T_{stl} unknown, in absence of other details, an interval in which T_{stl} can vary has to be defined, on the basis of the typical and reasonable PLL settling times values. Considering that the settling time can be reasonably tuned in an interval going from 100 ms to 600 ms, the initial guess of the nominal synchronization loop $\bar{G}_{syn}(s)$ is assumed with $T_{stl} = 350$ ms, exactly in the middle of the interval.

The initial definition of the uncertainty by means of w_{syn} and $\Delta_{syn}(s)$ must be done under several consideration. Firstly, the uncertainty must cover all the possible values of the settling times T_{stl} in the defined interval, as shown in Fig. 3(a). Secondly, the uncertainty has to be higher in the high frequency range, since the possible presence of a positive-sequence detector or harmonic rejection algorithms affect the high frequencies more than the low frequencies [25], [27], [28]. In the light of this, $\Delta_{syn}(s)$ is chosen as a high-pass filter to give medium uncertainty at low frequency and high uncertainty at high frequency:

$$\Delta_{syn}(s) = \frac{s + \tau_{z,syn}}{s + \tau_{p,syn}} \tag{6}$$

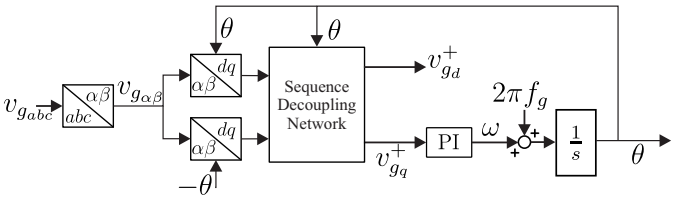


Fig. 4: The schematic of the Decoupled Double Synchronous Reference Frame PLL (DDSRF-PLL) [28].

The coefficient w_{syn} is initialized to 0.6 (60% uncertainty on the nominal plant $\bar{G}_{syn}(s)$). If the so-defined $\Delta_{syn}(s)$ and w_{syn} are not able to ensure grid robust stability, then they will be refined in section VI to a value able to warranty grid stability. The resulting synchronization loop transfer function $G_{syn}(s)$ is shown in Fig. 3(a). The blue curve in Fig. 3(a) is the initial guess $\bar{G}_{syn}(s)$ of the synchronization loop, while the white region in the Bode plot is the admissible region where the frequency response of the real implementation $G_{syn}(s)$ must lay.

B. DC voltage control loop $G_{dvc}(s)$

The implementation of the dc-link voltage controller generally includes an integral action, to ensure zero steady-state error. As nominal plant, therefore, a PI controller is assumed:

$$\begin{cases} i^*_{g_d} & = \bar{G}_{dvc}(s)\left(v^*_{dc} - v_{dc}\right) \\ \bar{G}_{dvc}(s) & = \dfrac{K_{p,_{dc}}s + K_{i,_{dc}}}{s} \end{cases} \tag{7}$$

The uncertainty on dc-link voltage controller is both on the parameters $K_{p,_{dc}}$ and $K_{i,_{dc}}$ and the implementation [25]. Since sequence detectors of harmonic compensation schemes are not commonly implemented in the dc voltage control, it is assumed that the uncertainty equally affects both low and high frequencies. Thus $\Delta_{dvc}(s)$ is chosen as a unitary gain constant along all frequencies.

$$\Delta_{dvc}(s) = 1 \tag{8}$$

The w_{dvc} is initialized to 0.3, covering bandwidth values between 15Hz and 40Hz, but the value can be eventually refined according to the procedure in section VI, in the case where the so-defined uncertainty is not able to warranties robust stability analyses, analogously to $w_{syn}\Delta_{syn}(s)$. With this initial definition of the uncertainty, the frequency response of $G_{dvc}(s)$ lays in the region depicted in white in Fig. 3(b).

C. AC voltage control loop $G_{avc}(s)$

Differently from the synchronization and dc voltage control, the design for the ac voltage control is constrained to the grid code, which generally impose a well-defined reactive current/power injection profile depending on the ac voltage [25]. Therefore, the ac voltage control is modeled in this paper with a constant and known gain:

$$\begin{cases} i^*_{g_q} & = \bar{G}_{avc}(s)\left(v^*_g - v_{g_d}\right) \\ \bar{G}_{avc}(s) & = K_{p,_{ac}} \end{cases} \tag{9}$$

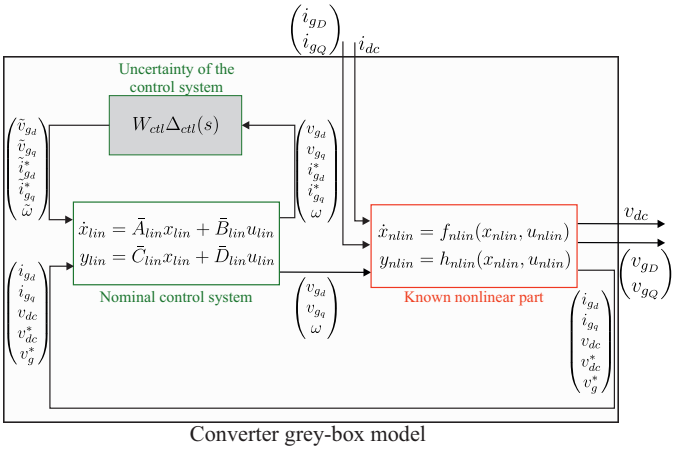


Fig. 5: The block interconnection procedure to embed the uncertainty in the proposed grey-box nonlinear converter model.

Having the knowledge of the grid code, it is reasonable to assume no uncertainty in the ac voltage control in the low frequency range, i.e. $\Delta_{avc}(0) = 0$. For the high frequency range, eventual variations in the controller implementation can be taken into consideration. $\Delta_{avc}(s)$ assumes thus the form of a high-pass filter with a zero in 0.

$$\Delta_{avc}(s) = \frac{s}{s + \tau_{p,avc}} \tag{10}$$

With the initial setting $w_{avc} = 0.3$, the resulting transfer function lays in the white region of the Bode plot shown in Fig. 3(c).

D. Inner current loop $G_{ci}(s)$

The current loop has generally a faster response than the outer loops, and contemplates various implementation strategies also with the presence of negative sequence and harmonic compensation schemes [25]. Therefore, its uncertainty is much higher in high frequency range than in low frequencies. A standard current loop implementation is the PI (or PR) controller tuned with technical optimum, which is modeled in this paper in the dq frame as a Multi-Input Multi-Output (MIMO) system according to the strategy described in [3]:

$$\begin{cases} v_g & = \bar{G}_{ci}(s)\begin{pmatrix} i^*_g \\ i_g \end{pmatrix} \\ \bar{G}_{ci}(s) & = \begin{pmatrix} \dfrac{K_p^2 s + K_p K_i}{L_f s^2 + K_p s} I_{2\times 2} & -\dfrac{K_p s + K_i}{s} I_{2\times 2} \end{pmatrix} \end{cases} \tag{11}$$

where $I_{2\times 2}$ is the identity matrix of dimension 2. For the uncertain model the uncertainty $\Delta_{ci}(s)$ is chosen as a MIMO high-pass filter in the form:

$$\Delta_{ci}(s) = \begin{pmatrix} \frac{s+\tau_{z,ci}}{s+\tau_{p,ci}} & 0 \\ 0 & \frac{s+\tau_{z,ci}}{s+\tau_{p,ci}} \end{pmatrix} \tag{12}$$

where $\tau_{z,ci}$ and $\tau_{p,ci}$ are chosen in order to give low uncertainty in the low frequency range. The scalar w_{ci} is initialized to the matrix $0.3I_{2\times 2}$ and the resulting uncertain transfer functions are shown in Fig. 3(d).

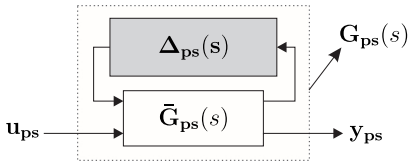


Fig. 6: The proposed MIMO uncertain linearized closed-loop system of the converter-dominated grid.

IV. Nonlinear Grey-Box State-Space Model of Converter-dominated Grids

The uncertain transfer functions of the control loops, defined in the previous section and shown in Fig. 3, constitute the *unknown control system model*, represented with the dashed green rectangle in Fig. 2. The latter is mathematically translated into the interconnection of the two blocks *nominal control system* and *uncertainty of the control system* in Fig. 5, as conventionally done in robust control theory to include multiplicative uncertainties in the form (4) by means of Linear Fractional Transformation [32]–[34]. The uncertainty of the control system, shown in Fig. 5, is defined as a 5×5 MIMO transfer matrix $\Delta_{ctl}(s)$ concatenating the uncertainties of the individual control loops, premultiplied by a weighting matrix W_{ctl} (note that w_{ci} and $\Delta_{ci}(s)$ are 2×2 matrices, as described in (12)), respectively defined as:

$$\Delta_{ctl}(s) = diag\left\{\Delta_{ci}(s), \Delta_{dvc}(s), \Delta_{avc}(s), \Delta_{syn}(s)\right\} \quad (13)$$

$$W_{ctl} = diag\left\{w_{ci}, w_{dvc}, w_{avc}, w_{syn}\right\} \quad (14)$$

where $diag$ is the diagonal block concatenation.

The remaining blocks of the converter model in Fig. 2 that lay outside the dashed green rectangle (dc converter side, rotation matrices and frequency/angle differential relationship) are embedded into the block *known nonlinear part*, shown in Fig. 5 with red outline. The resulting converter model in Fig. 5 is favourable for robust stability analyses thanks to the following characteristics:

- The uncertainty $W_{ctl}\Delta_{ctl}(s)$ is interconnected with a fully linear subsystem, according to Fig. 5, facilitating the application of robust control theory [32].
- Despite the converter model is nonlinear, its nonlinear part can be isolated and condensed into the red subsystem, which is not directly affected by model uncertainty.
- The converter model in Fig. 5 can be used as a building block for nonlinear converter-dominated grids models [3].

A. Interconnection of power converters models with the power network

A power grid with 3 VSCs with unknown control systems, as in Fig. 1(b), is considered. The VSC1 and VSC2 are manufactured by the same vendor, denominated Vendor #1, and have identical control systems; the VSC3 is produced by a Vendor #2 with a different control system. Vendor #1 and Vendor #2 keep their control systems confidential, however, they both declare to use a grid-following control and to comply with the interoperability specifications defined by the system operator in the form of frequency-domain boundaries on the control loops, as illustrated in Fig. 3 and discussed in section III. Therefore, the grey-box model in Fig. 5 is used to mathematically model both the VSC from Vendor #1 and #2 in Fig. 1(b). The uncertainty on the control system implemented by Vendor #1 is denominated $\Delta_{ctl_1}(s)$, while the uncertainty of the Vendor #2 is $\Delta_{ctl_2}(s)$. The fact that $\Delta_{ctl_1}(s)$ and $\Delta_{ctl_2}(s)$ are distinct means that the control systems of Vendor #1 and Vendor #2 are different. Nevertheless, the frequency weighting of $\Delta_{ctl_1}(s)$ and $\Delta_{ctl_2}(s)$ is chosen identical, according to Fig. 3. For the power grid modeling, the two uncertainties are concatenated in the matrix $\mathbf{\Delta_{ps}(s)} := diag\left\{W_{ctl_1}\Delta_{ctl_1}(s), W_{ctl_2}\Delta_{ctl_2}(s)\right\}$.

The nonlinear DAE model of the grid in Fig. 1(b) is built according to the general procedure in [3], briefly recalled in this subsection, extendable also to a higher number of converters and different power network configurations. The uncertainty $\mathbf{\Delta_{ps}(s)}$ will be added to the model at the end of the interconnection procedure, after the grid model linearization.

The n nonlinear VSCs nominal plant models (in this case $n = 3$) are concatenated in a column vector, considering the input i_{dc} of the VSC model represented in Fig. 5 to be $\frac{P^*}{v_{dc}}$, with P^* the amount of power provided by the DC energy sources in Fig. 1(b) (supposed constant):

$$\begin{pmatrix} \dot{x}_{vsc_1} \\ \dot{x}_{vsc_2} \\ \vdots \\ \dot{x}_{vsc_n} \end{pmatrix} = \begin{pmatrix} f_{vsc_1}(x_{vsc_1}, \frac{P_1^*}{v_{dc_1}}, i_{g,DQ_1}) \\ f_{vsc_2}(x_{vsc_2}, \frac{P_2^*}{v_{dc_2}}, i_{g,DQ_2}) \\ \vdots \\ f_{vsc_n}(x_{vsc_n}, \frac{P_n^*}{v_{dc_n}}, i_{g,DQ_n}) \end{pmatrix} \quad (15)$$

$$\begin{pmatrix} v_{g,DQ_1} \\ v_{g,DQ_2} \\ \vdots \\ v_{g,DQ_n} \end{pmatrix} = \begin{pmatrix} h_{vsc_1}(x_{vsc_1}, i_{g,DQ_1}) \\ h_{vsc_2}(x_{vsc_2}, i_{g,DQ_2}) \\ \vdots \\ h_{vsc_n}(x_{vsc_n}, i_{g,DQ_n}) \end{pmatrix} \quad (16)$$

The model of the n VSCs in (15) and (16) is then expressed in compact form using a bold notation for the vectorial variables, e.g. $\mathbf{v_g} = \begin{pmatrix} v_{g,DQ_1} & v_{g,DQ_2} & \cdots & v_{g,DQ_n} \end{pmatrix}^T$:

$$\begin{cases} \dot{\mathbf{x}}_{\mathbf{vsc}} & = \mathbf{f_{vsc}}(\mathbf{x_{vsc}}, \frac{\mathbf{P^*}}{\mathbf{v_{dc}}}, \mathbf{i_g}) \\ \mathbf{v_g} & = \mathbf{h_{vsc}}(\mathbf{x_{vsc}}, \mathbf{i_g}) \end{cases} \quad (17)$$

The main electric grid, represented with the voltage source e in Fig. 1(b), is modeled with the following nonlinear system, describing its voltage amplitude and frequency dynamics:

$$\Psi : \begin{pmatrix} E_g \\ \Delta\omega_g \end{pmatrix} \rightarrow \begin{pmatrix} e_D \\ e_Q \end{pmatrix} = \begin{cases} \dot{\delta}_g & = \Delta\omega_g \\ e_D & = E_g \cos\delta_g \\ e_Q & = E_g \sin\delta_g \end{cases} \quad (18)$$

The modeling of the power network uses differential equations for representing the electromagnetic dynamics of the distribution lines, given its coupling with the fast time-scale control of power converters:

$$\begin{cases} \dot{\mathbf{x}}_{\mathbf{net}} & = \mathbf{A_{net}}\mathbf{x_{net}} + \mathbf{B_{net}}\mathbf{v_g} + \mathbf{F_{net}}\Psi(E_g, \Delta\omega_g) \\ \mathbf{i_g} & = \mathbf{C_{net}}\mathbf{x_{net}} \end{cases} \quad (19)$$

The interconnection between the converters models and power network model assumes the form:

$$\begin{pmatrix}\dot{\mathbf{x}}_{\mathbf{vsc}}\\ \dot{\mathbf{x}}_{\mathbf{net}}\end{pmatrix} = \begin{pmatrix}\mathbf{f}_{\mathbf{vsc}}\left(\mathbf{x}_{\mathbf{vsc}}, \frac{\mathbf{P}^*}{\mathbf{v_{dc}}}, \mathbf{C}_{\mathbf{net}}\mathbf{x}_{\mathbf{net}}\right)\\ \mathbf{A}_{\mathbf{net}}\mathbf{x}_{\mathbf{net}} + \mathbf{B}_{\mathbf{net}}\mathbf{h}(\mathbf{x}_{\mathbf{vsc}}, \mathbf{C}_{\mathbf{net}}\mathbf{x}_{\mathbf{net}}) + \mathbf{F}_{\mathbf{net}}\Psi(E_g, \Delta\omega_g)\end{pmatrix} \tag{20}$$

By defining $\mathbf{x_{ps}} = \begin{pmatrix}\mathbf{x_{vsc}}^T & \mathbf{x_{net}}^T\end{pmatrix}^T$ and $\mathbf{u_{ps}} = \begin{pmatrix}\mathbf{P}^* & E_g & \Delta\omega_g\end{pmatrix}^T$, the nominal grid model (20) can be written in compact form as:

$$\begin{cases}\dot{\mathbf{x}}_{\mathbf{ps}} &= \mathbf{f}_{\mathbf{ps}}\left(\mathbf{x}_{\mathbf{ps}}, \mathbf{u}_{\mathbf{ps}}\right)\\ \mathbf{y}_{\mathbf{ps}} &= \mathbf{h}_{\mathbf{ps}}\left(\mathbf{x}_{\mathbf{ps}}, \mathbf{u}_{\mathbf{ps}}\right)\end{cases} \tag{21}$$

where $\mathbf{y_{ps}}$ contains the variables used for the power flow analysis (voltage, angle, active and reactive power of each converter), reported in Table IV. The linearization of (21) requires the computation of the static equilibrium point $\mathbf{x_{ps}}_e$ by solving through Newton-Raphson the equation:

$$\mathbf{f_{ps}}(\mathbf{x_{ps}}, \mathbf{u_{ps}}_e) = 0 \tag{22}$$

for a defined and constant input $\mathbf{u_{ps}}_e$ [3]. Following the considerations in section III, the total uncertainty $\mathbf{\Delta_{ps}(s)}$ does not play any role in the *static* converters and grid behaviour, but it influences only their *transient* dynamics. Thus, the uncertainty is considered null ($\mathbf{\Delta_{ps}(s)} = 0$) in the nonlinear static model (22) used solely for the equilibrium point computation, and is included in the grid model only *after* its linearization, as it will be explained in this paragraph.

The computation of $\mathbf{x_{ps}}_e$ through the solution of (22) is a generalized power flow problem [3], [22]. Once $\mathbf{x_{ps}}_e$ is computed, the steady-state power flow variables in $\mathbf{y_{ps}}_e$ are obtained through the nonlinear algebraic function $\mathbf{y_{ps}}_e = \mathbf{h_{ps}}\left(\mathbf{x_{ps}}_e, \mathbf{u_{ps}}_e\right)$. The resulting $\mathbf{y_{ps}}_e$ for a single converter connected to the grid, as in Fig. 1(a), is reported in Table III under different SCR values and grid voltage levels E_g. The computed $\mathbf{y_{ps}}_e$ for the power grid in Fig. 1(b) is instead presented in Table IV, under two different operating points determined by different power injections, denominated A (partial power injection) and B (full power injection). The model (21) is then linearized around the computed $\mathbf{x_{ps}}_e$, obtaining:

$$\begin{cases}\dot{\mathbf{x}}_{\mathbf{ps}} &= \bar{\mathbf{A}}_{\mathbf{ps}}\mathbf{x}_{\mathbf{ps}} + \bar{\mathbf{B}}_{\mathbf{ps}}\mathbf{u}_{\mathbf{ps}}\\ \mathbf{y}_{\mathbf{ps}} &= \bar{\mathbf{C}}_{\mathbf{ps}}\mathbf{x}_{\mathbf{ps}} + \bar{\mathbf{D}}_{\mathbf{ps}}\mathbf{u}_{\mathbf{ps}}\end{cases} \tag{23}$$

The linearized nominal model of the power grid (23) is then expressed as a transfer function $\bar{\mathbf{G}}_{\mathbf{ps}}(s)$, and interconnected with the total power system uncertainty $\mathbf{\Delta_{ps}(s)}$, yielding to the uncertain power grid model $\mathbf{G_{ps}}(s)$, as in Fig. 6:

$$\mathbf{G_{ps}} = \bar{\mathbf{G}}_{\mathbf{ps,22}} + \bar{\mathbf{G}}_{\mathbf{ps,21}}\mathbf{\Delta_{ps}}\left(\mathbf{I} - \bar{\mathbf{G}}_{\mathbf{ps,11}}\mathbf{\Delta_{ps}}\right)^{-1}\bar{\mathbf{G}}_{\mathbf{ps,12}} \tag{24}$$

with $\mathbf{I}$ identity matrix and $\bar{\mathbf{G}}_{\mathbf{ps,ii}}$, $\mathbf{i} = 1, 2$ being the sub-blocks of $\bar{\mathbf{G}}_{ps}(s)$ [32].

V. Robust Stability Analysis

The first step for the proposed robust stability analysis is the computation of the power grid stability margin α_{min} in presence of uncertain converter models, defined as:

$$\alpha_{min} := \min_{\mathbf{\Delta_{ps}}}\left[\bar{\sigma}(\mathbf{\Delta_{ps}(s)}) : \det\left(\mathbf{I} - \bar{\mathbf{G}}_{\mathbf{ps,21}}\mathbf{\Delta_{ps}}\right) = 0\right] \tag{25}$$

being $\bar{\sigma}(\mathbf{\Delta_{ps}(s)})$ the maximum singular value of $\mathbf{\Delta_{ps}(s)}$, which belongs the defined uncertainty set [32]. The solution of (25) is computed through the command `robstab()` in Matlab®.

TABLE II: Parameters considered for the robust stability analysis in single converter case.

Grid and converter parameters	**Values**
Line-to-line grid voltage (V)	690
Short Circuit Ratio	variable
R/X ratio	0.3
DC-link capacitor (mF)	22
Converters Control Parameters	**VSC**
Active power reference P^* (MW)	2
Switching frequency (kHz)	2
DC-link voltage reference (V)	1100
DC voltage controller bandwidth	unknown
AC voltage controller tuning	unknown
PLL settling times (s)	unknown

TABLE III: Equilibrium point of the single converter under different SCRs and grid voltage E_g steady-state values, obtained by (22).

E_g	Steady-state var. in $\mathbf{y_{ps}}_e$	SCR=2	SCR=3	SCR=4	SCR=5
1 p.u.	$\|v_g\|$ (p.u.)	1.00	1.01	1.02	1.02
1 p.u.	δ (deg)	14	9.35	6.96	5.55
1 p.u.	P (p.u.)	2	2	2	2
1 p.u.	Q (p.u.)	-0.05	-0.14	-0.17	-0.17
0.95 p.u.	$\|v_g\|$ (p.u.)	0.98	0.99	0.99	0.99
0.95 p.u.	δ (deg)	15	9.82	7.30	5.82
0.95 p.u.	P (p.u.)	2	2	2	2
0.95 p.u.	Q (p.u.)	0.09	0.03	0.03	0.05
0.9 p.u.	$\|v_g\|$ (p.u.)	0.96	0.97	0.96	0.96
0.9 p.u.	δ (deg)	16	10.3	7.69	6.13
0.9 p.u.	P (p.u.)	2	2	2	2
0.9 p.u.	Q (p.u.)	0.23	0.20	0.22	0.25
1.05 p.u.	$\|v_g\|$ (p.u.)	1.02	1.04	1.04	1.05
1.05 p.u.	δ (deg)	13	8.93	6.65	5.31
1.05 p.u.	P (p.u.)	2	2	2	2
1.05 p.u.	Q (p.u.)	-0.20	-0.32	-0.37	-0.40

A. Robust stability margin of a single converter

Firstly, the case of single grid following converter (see Fig. 1(a)) connected to an ac grid is considered, with the parameters declared in Table II. In this case, the control system uncertainty is $\mathbf{\Delta_{ps}}(s) = W_{ctl}\Delta_{ctl}(s)$, as in Fig. 5. Fig. 7(a) shows the stability margin (expressed in percentage) of the converter model linearized around different equivalent grid voltage levels E_g, under different SCRs.

The interpretation of the stability margin α_{min} defined in (25) and shown in Fig. 7(a) is the following. Considering the case of SCR=4, a robust stability margin of 118% for $E_g = 1$ p.u. means that the *smallest* perturbation (in terms of $\mathcal{H}_\infty$ norm) able to destabilize the system has an $\mathcal{H}_\infty$ norm of 118% of the maximum defined admissible perturbation $W_{ctl}\Delta_{ctl}(s)$ in Fig. 5. This means that it does not exist a destabilizing perturbation in the defined range of uncertainty, and the power grid can be considered stable for all the the possible control tuning included in the uncertainty $W_{ctl}\Delta_{ctl}(s)$ in Fig. 5.

Differently, the case with SCR=2 and $E_g = 0.9$ p.u. in Fig. 7(a) gives a stability margin of 83%. This means, there exist

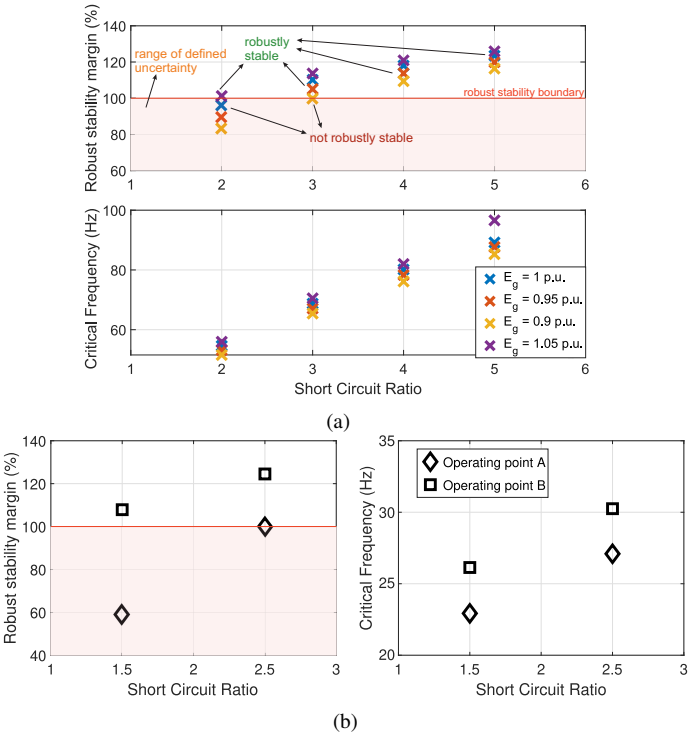


Fig. 7: Robust stability analysis showing the stability margin and the critical frequency. (a) Single converter case under different SCR values. (b) Multi-converter scenario with different operating points.

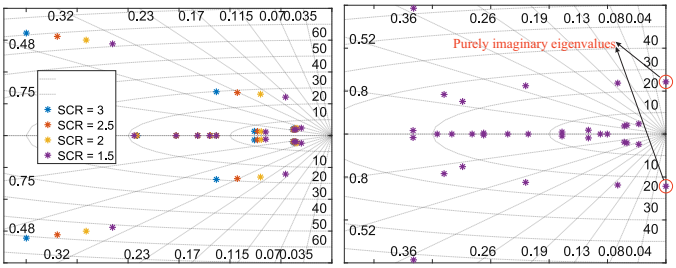


Fig. 8: The eigenvalue analyses of the converter-dominated grid working at operating point A. (a) Nominal implementation for the control loops under different SCRs. (b) Destabilizing control system, under SCR=1.5.

at least one destabilizing perturbation with an $\mathcal{H}_\infty$ norm equal or higher than 83% of the maximum admissible perturbation $W_{ctl}\Delta_{ctl}(s)$. Clearly, in this case the system in not robustly stable, since at least one unstable control implementation exists in the defined range of uncertainty.

Overall, the assessment of the robust stability through the robust stability margin in Fig. 7(a) follows this intuitive rule:

Robust Stability Assesment:. *Given a range of uncertainty for the control system implementation, defined by a bounded multiplicative uncertainty, the system is robustly stable if the robust stability margin is higher than 1 (i.e. higher than 100%), and not robustly stable if the robust stability margin is lower than 1 (i.e. lower than 100%).*

According to Fig. 7(a), the stability margin of the grid

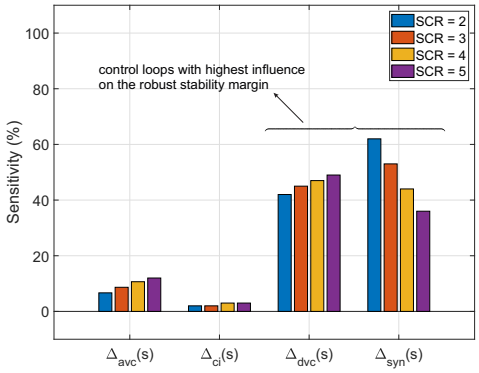


Fig. 9: The sensitivity of the robust stability margin with respect to the individual control loops.

TABLE IV: Power flow analysis by solving (22) under different grid operating points and Short Circuit Ratio values

Op. Point	Op. Point A		Op. Point B	
Var. in $\mathbf{y_{PS}}_e$	SCR=1.5	SCR=2.5	SCR=1.5	SCR=2.5
$v_{g,1}$ (p.u.)	0.95	0.97	0.93	0.97
δ_1 (deg)	19.7	11.43	19.82	11.35
P_1 (MW)	**0.50**	**0.50**	**1.00**	**1.00**
Q_1 (MVar)	0.23	0.13	0.28	0.15
$v_{g,2}$ (p.u.)	1.01	1.02	1.01	1.03
δ_2 (deg)	20.28	12.19	20.64	12.43
P_2 (MW)	**0.50**	**0.50**	**0.50**	**0.50**
Q_2 (MVar)	-0.03	-0.12	-0.04	-0.17
$v_{g,3}$ (p.u.)	1.04	1.06	1.06	1.08
δ_3 (deg)	20.69	12.69	21.24	13.14
P_3 (MW)	**1.00**	**1.00**	**1.50**	**1.50**
Q_3 (MVar)	-0.21	-0.30	-0.31	-0.44

following converter in Fig. 1(a) decreases with lower SCR values and lower grid voltage levels E_g and vice versa, in accordance with the literature [18]. It is important to notice in Fig. 7(a) that the stability margin varies to a high extent depending on the steady-state equivalent grid voltage E_g. Conventional impedance models are not able to capture this dependency on E_g, since they are generally identified in nominal conditions ($E_g = 1$ p.u.) and are accurate only in that condition. The analysis in Fig. 7(a) clearly demonstrates the potential inaccuracies introduced by small-signal models, and justify the need of nonlinear converter models [3], [8].

The frequency around which the instability occurs, in case of a particular control tuning of the power converters, is denominated *critical frequency* and is computed through robust analysis tools, shown in Fig. 7. The critical frequency turns very useful when a damping action for a grid resonance instability has to be designed without having full knowledge of the grid model.

B. Robust stability analysis for assessing converters interoperability

The power grid in Fig. 1(b) is considered and modeled with the grey-box modeling strategy illustrated in Fig. 6 derived in section IV. The two grid operating points (denominated A and B in Table IV) are considered, determined by different levels of active power injection for the three VSCs, highlighted in bold in Table IV. The resulting grid equilibrium points, computed through (22), are reported in Table IV. The power grid model is

TABLE V: Implementation and tuning of the converter control system in the HIL results: initial guess compared with the actual control implementation of the two vendors, considering two different tuning scenarios (A and B).

Control Loop	Initial guess for nominal plant	Control tuning scenario A		Control tuning scenario B	
		Vendor #1	Vendor #2	Vendor #1	Vendor #2
G_{syn} implem. G_{syn} tuning	SRF-PLL $T_{stl} = 350$ms	SRF-PLL $T_{STL} = 420$ms	DDSRF-PLL $T_{STL} = 380$ms, $\omega_{DF} = 35$Hz	SRF-PLL $T_{STL} = 190$ms	DDSRF-PLL $T_{STL} = 260$ms, $\omega_{DF} = 30$Hz
G_{dvc} implem. G_{dvc} tuning	PI with $(v_{dc}^{*} - v_{dc})$ BW$_{dvc}$ = 27.5Hz	PI with $(v_{dc}^{2*} - v_{dc}^{2})$ BW$_{DVC}$ = 29Hz	PI with $(v_{dc}^{*} - v_{dc})$ BW$_{DVC}$ = 26Hz	PI with $(v_{dc}^{2*} - v_{dc}^{2})$ BW$_{DVC}$ = 20Hz	PI with $(v_{dc}^{*} - v_{dc})$ BW$_{DVC}$ = 19Hz
G_{avc} implem. G_{avc} tuning	Droop $I_q - V$ $K_{p,ac} = 10$	Droop $Q - V$ $K_{p,AC} = 5630$	Droop $Q - V$ $K_{p,AC} = 5635$	Droop $Q - V$ $K_{p,AC} = 5625$	Droop $Q - V$ $K_{p,AC} = 5625$
G_{ci} implem. G_{ci} tuning	PI controller BW$_{ci}$ = 190Hz	PR controller BW$_{CI}$ = 215Hz	PI controller BW$_{CI}$ = 200Hz	PR controller BW$_{CI}$ = 185Hz	PI controller BW$_{CI}$ = 175Hz

linearized around the two different equilibrium points A and B, and the obtained robust stability margins α_{min} are computed and illustrated in Fig. 7(b). The results clearly highlight that the stability margins highly depend on the grid equilibrium point, and assess that operating point A is not robustly stable and operating point B is robustly stable. The fact that there exist operating points and SCR values for which the grid in Fig. 1(b) is not robustly stable implies that the *initial* tuning boundaries defined in section III and illustrated in Fig. 3 are *not* the interoperability specifications for the considered grid. In other words, the defined uncertainty $\mathbf{\Delta_{ps}(s)}$ is not adequate for warranting grid stability, and must be therefore redesigned.

VI. Determination of the Interoperability Specifications for Power Converters

The proposed method for designing a sufficiently large uncertainty $\mathbf{\Delta_{ps}(s)}$ which warranties converter interoperability and power grid stability contemplates three steps:

A) Analyse the stability of the *nominal* power grid model (i.e. $\mathbf{\Delta_{ps}(s)} = 0$) by means of eigenvalue analysis, and find a stable and well-damped nominal plant.
B) Perform a sensitivity analysis for the robust stability margin to detect which control loops are likely to bring instability with the considered uncertainty $\mathbf{\Delta_{ps}(s)}$.
C) Perform the destabilizing perturbation analysis on the critical control loops, to find which values of $\mathbf{\Delta_{ps}(s)}$ are able to destabilize the system. Then, redesign the uncertainty in order to avoid unstable plants inside the defined range.

A. Stability analysis of the nominal plant

In Fig. 8(a), the eigenvalues plot considering the nominal plant of the grid (i.e. with $\mathbf{\Delta_{ps}(s)} = 0$) are depicted. The eigenvalues are stable for all the considered SCR values, and move progressively to the right half plane as the SCR decreases, in accordance with the literature [4]. The procedure described in this section for determining interoperability specifications necessitate the stability of the nominal plant. With stable nominal system, indeed, when reducing the amount of uncertainties $w_{syn}\Delta_{syn}(s)$, $w_{dvc}\Delta_{dvc}(s)$, $w_{avc}\Delta_{avc}(s)$, $w_{ci}\Delta_{ci}(s)$, at some point the uncertain power grid model becomes robustly stable.

B. Sensitivity analysis of the control loops

The sensitivity analysis is aimed to find *which* control loop is the most critical for the robust stability and consequently need a refinement of its uncertainty to have stricter tuning boundaries. In Fig. 9, the sensitivity of the stability margin with respect to variations in the control loops tuning is computed through the command `robstab()` under different SCRs. The obtained values indicate how much a variation in the normalized perturbations of each control loop influence the grid stability margin, in percentage. It is found that, in the considered scenario, the design and tuning of the current controller $G_{ci}(s)$ as well as ac voltage controller $G_{avc}(s)$ have a relatively low influence on the stability margin. On the other hand, the tuning of the synchronization loop $G_{syn}(s)$ has the highest influence on the stability margin under a grid SCR lower than 3, where the grid is not robustly stable according to Fig. 7(b), while the dc voltage controller $G_{dvc}(s)$ has the highest influence on the stability margin with SCR of 4 or higher, where the grid is instead robustly stable.

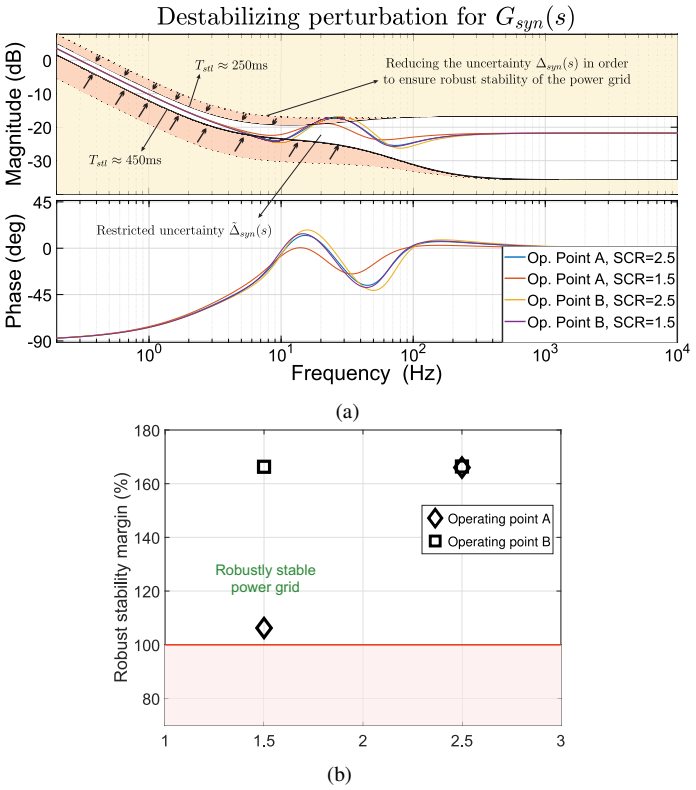


Fig. 10: (a) The destabilizing perturbation analysis for the synchronization loop $G_{syn}(s)$, which refinement of the uncertainty $\Delta_{syn}(s)$ to ensure robust grid stability. (b) The robust stability analysis considering the refined uncertainty $\tilde{\Delta}_{syn}(s)$.

C. Destabilizing tuning of the control loops and refinement of the uncertainty design

According to the sensitivity analysis in Fig. 9, the synchronization loop is likely to be the main responsible of the not robustly stable power grid at low SCRs, revealed by Fig. 7(b). This means that the PLL tuning boundaries, i.e. the uncertainty $w_{syn}\Delta_{syn}(s)$ depicted in Fig. 3(a), may be too large and permissive, and some unstable plants exist inside the white region of the Bode plot. The uncertainty $w_{syn}\Delta_{syn}(s)$ must be thus redesigned and properly reduced.

By solving the minimization problem in (25), the smallest perturbation $\mathbf{\Delta_{ps}(s)}$ in an $\mathcal{H}_\infty$ sense which makes the power grid unstable can be found. Such value of the transfer function is denominated in this paper *destabilizing perturbation*, indicated with the hat notation, e.g. $\mathbf{\hat{\Delta}_{ps}(s)}$. By substituting the $\mathbf{\hat{\Delta}_{ps}(s)}$ into the overall grid model in Fig. 6 and computing the closed loop poles, indeed, a pair of purely imaginary poles is present as shown in Fig. 8(b). That indicates that the power grid reaches its stability limit exactly for $\mathbf{\Delta_{ps}(s)} = \mathbf{\hat{\Delta}_{ps}(s)}$, confirming the validity of the analyses in Fig. 7 and also demonstrating that robust analyses methodologies are accurate and with no conservatism [33].

The destabilizing perturbation of the synchronization loop $\hat{\Delta}_{syn}(s)$ is shown in Fig. 10(a) for the operating points A and B and different SCR values. In the case of Operating Point A and SCR equal to 1.5 (in orange), the destabilizing perturbation lays inside the initially defined uncertainty (indicated with dashed lines in Fig. 10(a)), making the grid not robustly stable, in accordance with Fig. 7(b). Thus, the uncertainty boundaries of $w_{syn}\Delta_{syn}(s)$ are reduced (in this case, only in the low frequency) until the destabilizing perturbation lays outside the uncertainty range, as shown in Fig. 10(a). Reducing such boundaries means that *stricter* tuning guidelines are imposed to converter vendors for the tuning of the PLL.

At this point, a new robust stability analysis considering the reduced uncertainty on the PLL, indicated with the tilde notation $\tilde{w}_{syn}\tilde{\Delta}_{syn}(s)$, must be performed. The analysis, shown in Fig. 10(b), confirm that the reduced uncertainty on the PLL $\tilde{w}_{syn}\tilde{\Delta}_{syn}(s)$ as in Fig. 10(a) is sufficient to ensure robust stability of the power grid. Thus, the resulting control system uncertainty $\mathbf{\tilde{\Delta}_{ps}(s)}$ with the new restricted $\tilde{w}_{syn}\tilde{\Delta}_{syn}(s)$ is the *interoperability specification* for the considered power grid, i.e. if all vendors tune their controllers inside the boundaries defined by $\mathbf{\tilde{\Delta}_{ps}(s)}$, the grid stability is warranted. $\tilde{w}_{syn}\tilde{\Delta}_{syn}(s)$ corresponds to a PLL implementation with a settling time T_{STL} between 250 ms and 450 ms, as shown in Fig. 10(a), with the possibility of implementing sequence decoupling networks or harmonic rejection schemes, thanks to the large uncertainty at high frequency range. The obtained parameters of $\mathbf{\tilde{\Delta}_{ps}(s)}$ which define the interoperability specifications for the converters in the considered grid are reported in Table VI.

TABLE VI: Parameters of the uncertainty $\mathbf{\tilde{\Delta}_{ps}(s)}$ able to ensure converters interoperability.

Control loop	Parameters for the uncertainty
Synchronization loop	$\tau_{z,syn} = 59$, $\tau_{p,syn} = 237$, $w_{syn} = 0.6$
DC-link voltage loop	$\tau_{z,dvc} = 0.3$
AC voltage loop	$\tau_{p,avc} = 32$, $w_{avc} = 0.3$
Current loop	$\tau_{z,ci} = 200$, $\tau_{p,ci} = 1000$, $w_{ci} = 0.3$

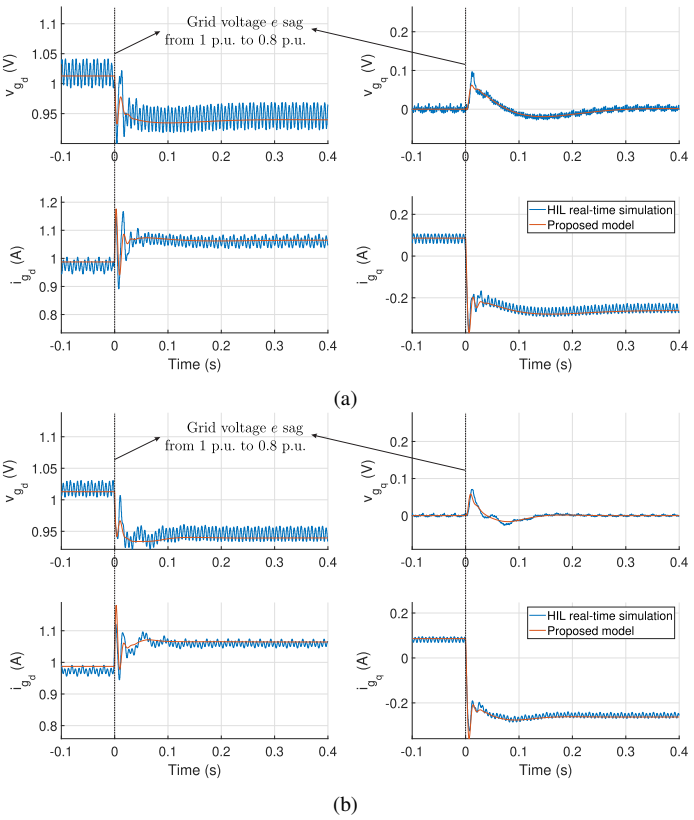


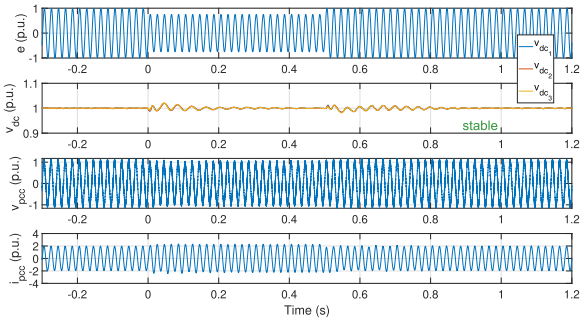
Fig. 11: The validation of the proposed converter model with respect to the two vendors described in Table V. (a) Vendor #1 in control tuning scenario A. (b) Vendor #2 in control tuning scenario B.

VII. Hardware-In-the-Loop Results

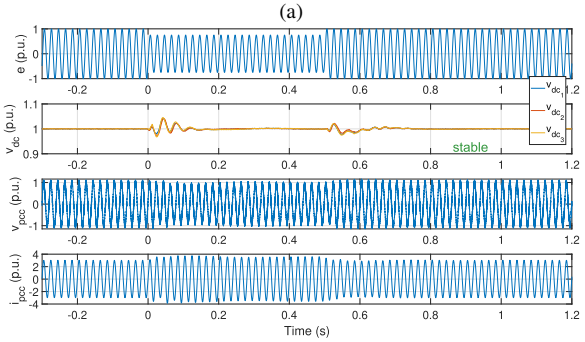
In order to validate all the previous analyses, the power grid in Fig. 1(b) is built and simulated in real-time on a Typhoon® HIL 402 device, considering a simulation step of 1 μs. The HIL results are aimed to firstly validate the proposed converter modeling methodology and afterwards to verify that, if both converter vendors comply with the interoperability specifications defined by $\mathbf{\tilde{\Delta}_{ps}(s)}$ described in Table VI, then the grid is robustly stable around different operating conditions.

A. Model Validation

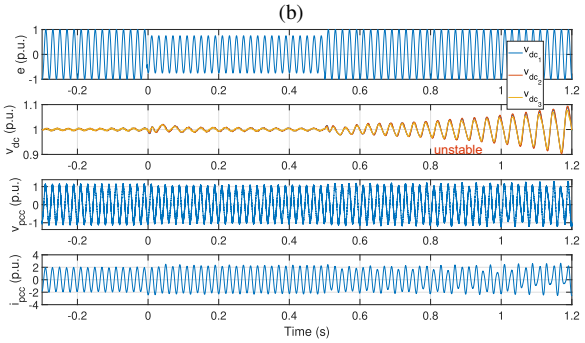
The key feature of proposed converter model, presented in section II and III, is its *generality*, which makes it suitable for the modeling of grid following converters with various control system implementation (e.g. different type of PLL). To demonstrate that, the power converters from Vendor #1 and #2 are built and simulated in real-time on the Typhoon® HIL device considering the control implementation and tuning described in Table V. The dynamic response of both converters subject to a grid voltage sag of 0.2 p.u. depth is compared with the simulation of the proposed converter models in Fig. 11, to validate the model accuracy. The DDSRF-PLL implemented by the Vendor #2, represented in Fig. 4, aims to decouple positive and negative sequence in the synchronization process,



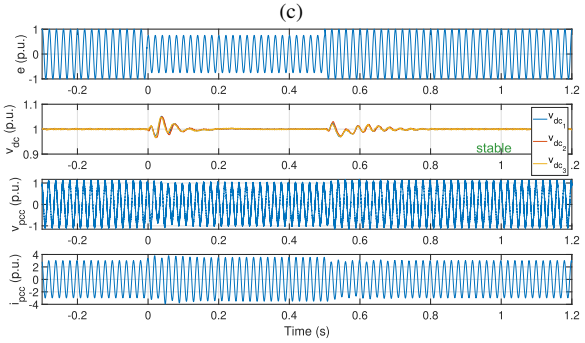
(a)



(b)



(c)

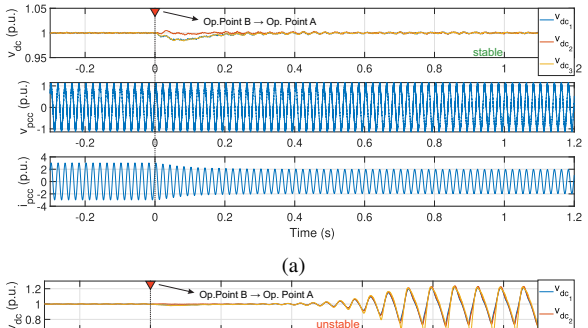


(d)

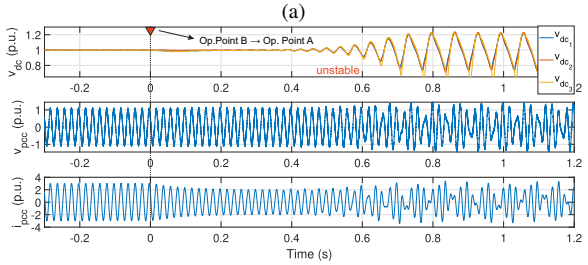
Fig. 12: The real-time HIL simulation of a voltage sag. (a) Operating point A with tuning A. (b) Operating point B with tuning A. (c) Operating point A with tuning B. (d) Operating point B with tuning B.

and is therefore modeled in the proposed converter model in Fig. 2 through a notch filter with central frequency at 100 Hz in the transfer function $G_{syn}(s)$, where the negative sequence appears in the dq frame.

The results of the validation both in Fig. 11(a) and in Fig. 11(b) confirms the accuracy of the proposed model for representing the control system implementation of both Vendor #1 and Vendor #2 which, with different transfer function



(a)



(b)

Fig. 13: The real-time HIL simulation of a sudden switch from operating point B to operating point A. (a) With tuning A. (b) With tuning B.

$G_{syn}(s)$, is able to accurately represent different kinds of implementation of the PLL in grid following control.

B. Robust Stability analysis validation

The HIL real-time simulation results in Figs. 12 and 13 are carried out considering the two different control tuning scenarios described in Table V and the two operating points analyzed in Table IV. In all tests of Figs. 12 and 13, it is considered SCR=1.5 and R/X ratio equal to 0.3.

The *real* control system (i.e. $G(s)$ in (4)) of the two vendors is deliberately designed in a substantially different manner and with different parameters from its *initial guess* (i.e. $\bar{G}(s)$ in (4)), for all control loops, as reported in Table V. Two control tuning scenarios are considered, denominated A and B in Table V. The real tuning parameters are declared with capital letters in the subscript (e.g. T_{STL}) in Table V, while parameters for the initial guess are indicated in lowercase (e.g. T_{stl}). In the tuning scenario A for the power converters, both Vendor #1 and #2 comply with the interoperability specifications defined by $\tilde{\mathbf{\Delta}}_{\mathbf{ps}}(\mathbf{s})$, derived in section VI. In the tuning scenario B, instead, Vendor #1 does not comply with the interoperability specifications due to the fast PLL tuning (see the tuning boundaries for the synchronization loop in Fig. 10(a)). Tuning B is only encompassed by the initial definition of the uncertainty $\mathbf{\Delta}_{\mathbf{ps}}(\mathbf{s})$ in Fig. 3, that has been however assessed to be *not robustly stable* by the analysis in Fig. 7(b).

Considering the synchronization loop $G_{syn}(s)$ in Fig. 3(a), the Vendor #1 implements it with a SRF-PLL, while the Vendor #2 uses a DDSRF-PLL with different tuning parameters, as described in Table V. Different values of T_{STL} shift the frequency response blue curve of Fig. 3(a) up (lower T_{STL}) or down (higher T_{STL}) in the Bode plot, with a slight modification of the curve shape. On the other hand, a different implementation of the synchronization loop, e.g. Decoupled Double SRF-PLL (DDSRF-PLL) with a decoupling filter with

bandwidth ω_{DF} of vendor #2 adds additional zeros and poles to $G_{syn}(s)$, modifying the curve shape in the medium-high frequency range, where the uncertainty $\tilde{\Delta}_{syn}(s)$ in Fig. 10(a) is large enough [25]. Similar considerations can be drawn for the other control loops in Table V, which differs both for implementation and tuning parameters between the two vendors and from the initial guess.

The considered grid depicted in Fig. 1(b) is subject to a 0.2 p.u. voltage sag of the duration of 0.5s, and the obtained HIL results are illustrated in Fig. 12. With tuning scenario A in Table V, complying with the interoperability specifications defined by $\tilde{\mathbf{\Delta}}_{\mathbf{ps}}(\mathbf{s})$ found in section VI, the stability is warranted and proven by the analysis in Fig. 10(b). As a matter of fact, the results of Figs. 12(a) and 12(b) show stable grid in both the considered operating conditions.

Different is the case of tuning scenario B, which is not compliant with the interoperability specifications, and originates an unstable behaviour when the grid is in operating point A, and a stable behaviour when the grid is in operating point B. As explained in section V, being not *robustly* stable does not directly imply that the grid always experiences instability, rather that there exists *no warranty* for the stability. In the case of operating point A, no warranty for the grid stability exists with the tuning scenario B, as assessed by the analysis in Fig. 7(b).

In the test of Fig. 13, a sudden switch from operating point B to operating point A, e.g. due to a sudden loss of some power generation, is simulated in HIL. The real-time simulation considering the tuning A (see Table V) is shown in Fig. 13(a). The grid in this case is able to remain stable in both operating point A and B, since the tuning scenario A complies with the interoperability specifications. The dc voltage v_{dc_2} from the VSC2 varies only slightly in Fig. 13(a), because the active power level P_2 of the VSCs remains constantly equal to 500 kW both in operating point A and B, according to Table IV. Conversely, the case of Fig. 13(b) considers the control tuning B (see Table V) which is not compliant with the interoperability specifications. In this case, the considered grid of Fig. 1(b) is stable in the operating point B and becomes unstable after the sudden switch to operating point A, which according to the analytical results in Fig. 7(b) is not robustly stable. The results in Fig. 13 provide a further confirmation of the validity of the robust stability analyses in section V and VI, and highlight above all the crucial importance of the grid operating points in grid stability studies, which is only taken into account by converter nonlinear models. The use of linear converter models for converter-dominated grids, therefore, may introduce inaccuracy in the stability analyses results.

VIII. Conclusions

In this paper, a methodology to analyze and warranty the stability of power electronics-based power systems without having a detailed knowledge of the converters control system is proposed. The advantages of the proposed approach over conventional solutions are here summarized:

- System operators are able to accurately analyze the stability of multi-vendor converter-dominated grids and find potential low-damped resonances without the need that converter manufacturers disclose details about the converters' control system.
- As a countermeasure to avoid grid instability, system operators can compute and provide interoperability specifications to converter vendors, in the forms of a set of tuning boundaries for the control loops able to warranty grid stability. The computation of these tuning boundaries is based on robust stability theory.
- The momentous impact of the converters' operating point on the grid stability has been clearly demonstrated, both through analytic and Hardware-In-the-Loop results. While conventional small-signal models are accurate only around one predefined equilibrium point and may lose accuracy at different equilibrium points, the proposed methodology based on nonlinear modeling of the converters and the power grid is accurate globally, and allow to detect and analyze hazardous operating conditions for the grid stability.
- Differently from black-box converter modeling, the proposed grey-box strategy allows to visualize the generic representation of the converters control system, and can provide by means of the sensitivity analysis and the destabilizing perturbation analysis precise tuning guidelines and interoperability specifications on the single control loops to ensure grid stability.

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